



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Fall 2022

Paper: Design and Analysis of Experiments (Theory)

Course Code: STAT-303



Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

- I. Partitioning the Sum of Squares in One-way ANOVA for Equal Sample Sizes.
- H. Efficiency of LS Design to RCB Design and CR Design
- III. ANCOVA
- IV. Scheffe's Test.
- V. Principles of Experimental Design
- VI. Orthogonal Polynomials

Answer the following questions:

(3x10=30)

- Q2. Derive the formulas for estimating Two missing observations in a Latin square design when the values are missing in different rows, different columns and in different treatments.
- Q3. Seven treatments arranged in six randomized complete blocks provided the following sum of squares and products:

S.O.V.	YY	XY	XX
Blocks	1200	600	200
Treatments	300	300	100
Error	1400	700	600

- i. Is the regression of Y on X significant at 0.05 level of significance?
- ii. Construct ANOVA and write the inference.

- Q4. A composite measure of screen quality was made on screen using four lacquer concentrations, four standing times, four acryloid concentrations (A,B,C,D) and four acetone concentrations ($\alpha, \beta, \gamma, \delta$). A Graeco Latin square design was used for the analysis purpose and the following results were obtained:

Source	Sum of Squares
Columns	25.25
Rows	85.25
Latin Letters	4.25
Greek Letters	26.25
Error	10.75
Total	151.75

Find the relative efficiency of this design with:

- i) Latin Square Design
- ii) Completely Randomized Design.